Regime-Aware Bid-Ask Spread Prediction: A Machine Learning Approach for SPX Market Microstructure

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Abstract

This study investigates the predictability of daily bid-ask spreads in the SP 500 index through a regime-aware machine learning framework. We combine Hidden Markov Models for market-regime identification with gradient boosting methods (XGBoost and LightGBM) to account for time-varying market conditions. These models are evaluated against two standard benchmarks—linear regression and a simple persistence model based on lag-1 relative spread.

Empirically, the machine learning models do not significantly outperform the lag-1 baseline. This result suggests that most of the predictable variation in daily SPX bid-ask spreads is captured by short-term autocorrelation, while the remaining fluctuations are largely dominated by noise. The findings highlight the inherent difficulty of forecasting high-frequency liquidity measures and the practical limits of model-driven spread prediction in real-world trading environments.

1 Introduction and Motivation

1.1 Background & Literature Review

The bid-ask spread is a fundamental component of transaction costs in financial markets, representing the compensation to market makers who provide liquidity (Glosten & Milgrom, 1985; Kyle, 1985). According to market microstructure theory, bid-ask spreads reflect three primary components: order processing costs, inventory holding costs, and adverse selection costs, arising from the asymmetric information among traders (Glosten & Milgrom, 1985; Kyle, 1985).

In the context of the S&P 500 index market, understanding and predicting spread dynamics has profound implications for portfolio managers, algorithmic traders, and regulatory bodies concerned with market quality and efficiency.

The relative importance of these components varies across market conditions. During periods of high volatility or market stress, spreads tend to widen as market makers face increased inventory risk and adverse selection problems. In contrast, in tranquil markets with low volatility and ample liquidity, spreads generally narrow (Glosten & Milgrom, 1985; Kyle, 1985).

From a methodological perspective, analyzing and forecasting spread dynamics requires careful treatment of time series properties. Covariance estimation that is robust to heteroskedasticity and autocorrelation can improve inference and model stability (Newey & West, 1987). Comparing predictive accuracy of alternative models is important for selecting the most effective forecasting approach (Diebold & Mariano, 1995). Moreover, modern machine learning techniques can be applied to quantify feature importance and understand drivers of spread dynamics (Lundberg & Lee, 2017).

1.2 Research Motivation

Despite extensive literature on bid-ask spread determinants, the predictive modeling of spreads remains challenging due to several factors:

Non-stationarity: Financial markets exhibit time-varying characteristics, with structural breaks corresponding to regime changes driven by macroeconomic conditions, monetary policy shifts, and market crises. Traditional models assuming stationarity may fail to capture these dynamics adequately.

High-dimensional feature space: Spread determination involves multiple interacting factors including price volatility, trading volume, order flow imbalances, and fundamental valuation metrics. Identifying the most relevant predictors and their time-varying relationships requires sophisticated statistical techniques.

Regime dependence: The relationship between spread determinants and realized spreads may differ fundamentally across market regimes. A predictor that strongly influences spreads during crisis periods may have negligible impact during normal market conditions.

Practical relevance: Accurate spread prediction enables more efficient execution strategies, improved transaction cost analysis, and better liquidity risk management. For institutional investors managing large portfolios, even marginal improvements in spread prediction can translate to substantial cost savings.

1.3 Research Objectives

This study addresses the aforementioned challenges through the following specific objectives:

1. Develop a regime identification framework using unsupervised learning techniques (K-Means clustering) and probabilistic state-space models (Hidden Markov Models).
2. Implement and compare multiple predictive models including linear regression baselines, gradient boosting machines (XGBoost, LightGBM), and persistence models.
3. Assess the incremental value of regime-aware features using both statistical significance tests and economic evaluation criteria.
4. Provide interpretable insights into spread determination mechanisms through SHAP value analysis and regime-specific performance decomposition.

1.4 Contribution

This research contributes to the market microstructure literature in several dimensions. Methodologically, we demonstrate the application of rolling window HMM estimation for regime detection in market data, addressing the challenge of regime persistence and stability. From an empirical standpoint, our findings quantify the economic significance of regime-aware modeling for spread prediction and identify specific market conditions under which traditional models fail.

2 Research Topic and Methodology

2.1 Data Description

Our analysis utilizes daily market data for the S&P 500 index spanning multiple years, obtained from Bloomberg. The dataset comprises:

Price and spread variables: PX_LAST (last traded price), PX_BID (best bid price), PX_ASK (best ask price), PX_MID (mid-price), and derived measures including absolute spread and relative spread (spread divided by mid-price).

Trading activity: PX_VOLUME (trading volume), transformed to log_volume to address skewness and heteroskedasticity.

Fundamental metrics: CUR_MKT_CAP (market capitalization), BEST_EPS (earnings per share), BEST_SALES (sales figures), and BEST_PE_RATIO (price-to-earnings ratio).

Return measures: Daily returns computed from price changes and lagged closing prices, as well as percentage changes.

2.2 Data Quality and Preprocessing

2.2.1 Quality Assurance

We implement comprehensive data quality checks to ensure analytical validity:

- Verification that bid prices never exceed ask prices
- Confirmation that spreads are non-negative
- Validation that mid-prices fall within the bid-ask bounds
- Detection and removal of extreme outliers using percentile-based filtering (retaining observations between the 1st and 99th percentiles)

After quality filtering, our dataset exhibits strong internal consistency with over 99% of observations satisfying fundamental microstructure constraints.

2.2.2 Feature Engineering

To avoid temporal data leakage, we construct all features using only information available at the time of prediction:

- **Volatility Measures:** Realized volatility computed over lagged returns for windows $t \in \{5, 21\}$, annualized by multiplying by $\sqrt{252}$:

$$\text{rv}_t = \sqrt{252} \cdot \sqrt{\frac{1}{t} \sum_{i=1}^{t} r_{t-i}^2}, \quad t \in \{5, 21\}.$$

- **Liquidity Proxies:** Amihud illiquidity measure:

$$\text{AMI}_t = \frac{|r_t|}{V_{t-1}}$$

- **Momentum Indicators:** Price and volume trends, computed as ratios of short-term to long-term moving averages:

$$\text{price_trend}_t = \frac{\text{MA}_5(P_{t-1})}{\text{MA}_{20}(P_{t-1})} - 1, \quad \text{volume_trend}_t = \frac{\text{MA}_5(V_{t-1})}{\text{MA}_{20}(V_{t-1})} - 1.$$

- **Autoregressive Features:** Lagged values of the relative spread for horizons $l \in \{1, 2, 3, 5\}$:

$$\text{spread_lag}_l(t) = \text{spread}_{t-l}, \quad l \in \{1, 2, 3, 5\}.$$

- **Fundamental Dynamics:** Changes in the price-to-earnings ratio:

$$\text{pe_diff}_t = \text{PE}_t - \text{PE}_{t-1}.$$

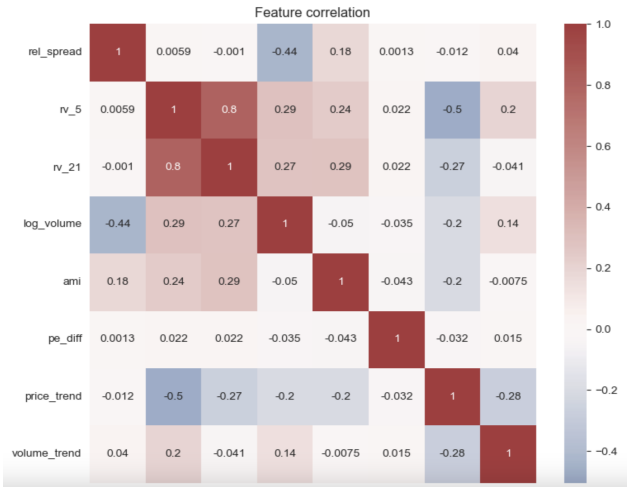


Figure 1: Feature Correlation Matrix

2.3 Regime Identification Framework

2.3.1 K-Means Clustering Analysis

We employ K-Means clustering to identify distinct market regimes based on standardized microstructure features. The clustering input includes realized volatility, log volume, relative spread, Amihud illiquidity, PE ratio changes, and price trends.

To determine the optimal number of clusters, we evaluate silhouette scores and inertia for $K \in \{2, 3, 4\}$. The silhouette score ranges from -1 to 1 with higher values indicating better-defined clusters. We also assess cluster quality through the Davies-Bouldin Index and Calinski-Harabasz Index.

2.3.2 Hidden Markov Model for Regime Detection

We implement a rolling window Hidden Markov Model approach with $K = 3$ hidden states and full covariance matrices. The HMM is defined by:

$$S_t \sim \text{Markov}(\pi, A) \tag{1}$$

$$X_t | S_t = k \sim \mathcal{N}(\mu_k, \Sigma_k) \tag{2}$$

Rather than fitting a single HMM to the entire sample, we estimate the model using a 756-day rolling window, refitting every 21 trading days. This approach allows regime definitions to evolve over time, adapting to structural changes in market microstructure.

2.4 Predictive Modeling Framework

2.4.1 Model Specifications

We implement and compare four distinct modeling approaches:

Persistence baseline: Uses the most recent observed spread as the prediction: $\hat{y}_{t+1} = y_t$

Linear regression: Ordinary least squares with all engineered features:

$$y_{t+1} = \beta_0 + \sum_{j=1}^{p} \beta_j x_{j,t} + \epsilon_{t+1} \tag{3}$$

XGBoost: Extreme Gradient Boosting with maximum tree depth of 3, learning rate of 0.05, and subsample fractions of 0.8.

LightGBM: Light Gradient Boosting Machine with comparable hyperparameters to XGBoost.

2.4.2 Time-Series Cross-Validation Protocol

We implement rigorous rolling window cross-validation: Each training iteration uses a 500-day training window, followed by a 50-day validation window for early stopping. After identifying the optimal number of iterations, we refit the model on the combined 550 days and make one-step-ahead predictions. The window advances by one day for each new prediction.

2.5 Evaluation Metrics

We assess model performance using:

Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4}$$

Mean Absolute Error:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{5}$$

R-squared:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6}$$

Diebold-Mariano test: For comparing forecast accuracy between competing models.

3 Findings and Discussion

3.1 Data Characteristics

The preprocessed dataset exhibits strong data quality, with 99.99% of observations satisfying standard microstructure consistency constraints. Descriptive statistics indicate substantial time-variation in relative spreads. The correlation matrix shows that relative spreads are moderately negatively correlated with log trading volume (≈ -0.44), while their correlation with short-horizon realized volatility is close to zero. In contrast, the two realized volatility measures (5-day and 21-day) are strongly correlated with each other (≈ 0.8), reflecting shared underlying volatility dynamics. The Amihud illiquidity measure displays only a weak positive correlation with spreads (≈ 0.18), suggesting that illiquidity and spreads capture partially distinct aspects of market frictions in this dataset. Additionally, the price trend variable exhibits a moderate negative correlation with spreads (≈ -0.5), indicating that stronger upward price movements tend to coincide with narrower spreads.

3.2 Regime Identification Results

3.2.1 K-Means Clustering

The silhouette score analysis reveals optimal clustering performance at $K = 4$ regimes:

Table 1: Cluster Characteristics

Cluster	rV_5	log_volume	rel_spread	ami	pe_diff	price_trend
1	0.4564	20.9222	0.00874	1.49e-11	0.06416	0.005196
2	0.1530	20.9005	0.00137	5.89e-12	0.02337	0.175711
3	0.1102	20.1648	0.01377	7.11e-12	-0.12220	-0.04594
4	0.1757	20.4148	0.01167	2.79e-11	0.00359	0.008094

Cluster 1 includes stocks with high volatility and high trading volume. They move significantly intraday and attract heavy trading, with moderate spreads indicating decent liquidity. Their price trend is mildly positive, and valuation signals are slightly favorable. These are active, attention-grabbing names with reasonable tradability.

Cluster 2 captures the most liquid and stable stocks. They have the tightest spreads, very high volume, and relatively low volatility. Their price trend is strongly upward, supported by a positive valuation differential. This cluster resembles large, highly traded names with strong market momentum.

Cluster 3 contains the least liquid stocks, with low volume and wide spreads. They show low volatility due to limited trading activity and have a negative price trend and unfavorable valuation signal. This cluster reflects illiquid or distressed names with weak momentum and fundamentals.

Cluster 4 represents stocks with mid-range behavior across all features. Their volume, spreads, and volatility are all moderate, and both trend and valuation signals are close to neutral. These are average, steady names without strong distinguishing characteristics.

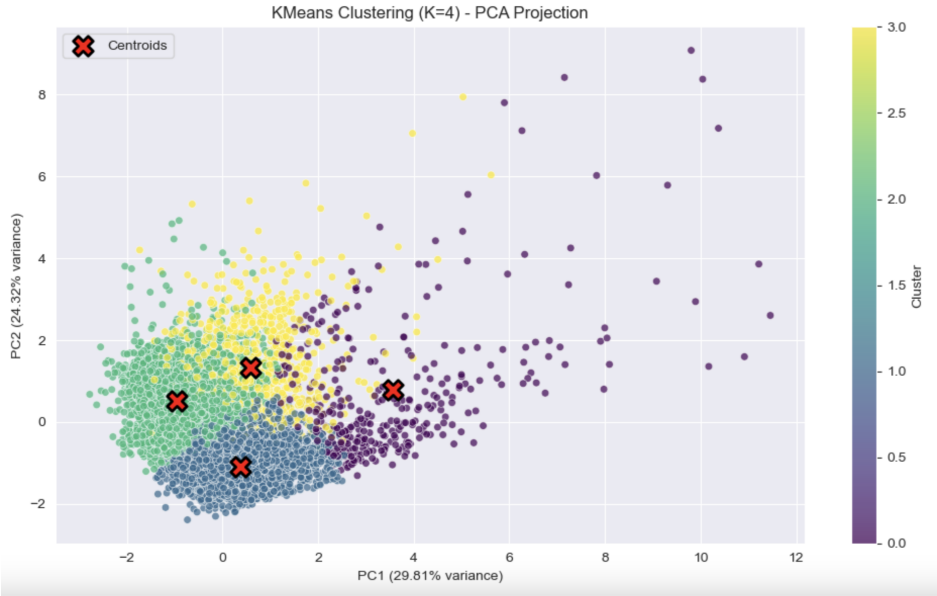


Figure 2: K-means with K=4

Although K = 4 produces the highest Silhouette score and inertia, indicating the best clustering performance numerically, the regime separation is not visually clear in the feature space, suggesting that the boundaries between regimes are somewhat overlapping.

3.2.2 Rolling Window HMM

The rolling window HMM successfully identifies time-varying regime states. Key findings:

Regime distribution: The two-state HMM yields a balanced regime structure, with Regime 0 accounting for approximately 55–65% of observations and Regime 1 for the remaining 35–45%.

Temporal stability: The transition probability matrix shows strong diagonal elements (0.9- 1), confirming that regimes persist over multiple days.

3.3 Out-of-Sample Predictive Performance

3.3.1 Aggregate Performance Metrics

The out-of-sample predictive performance of the evaluated models is summarized in Table 1 and visualized in Figure 3.

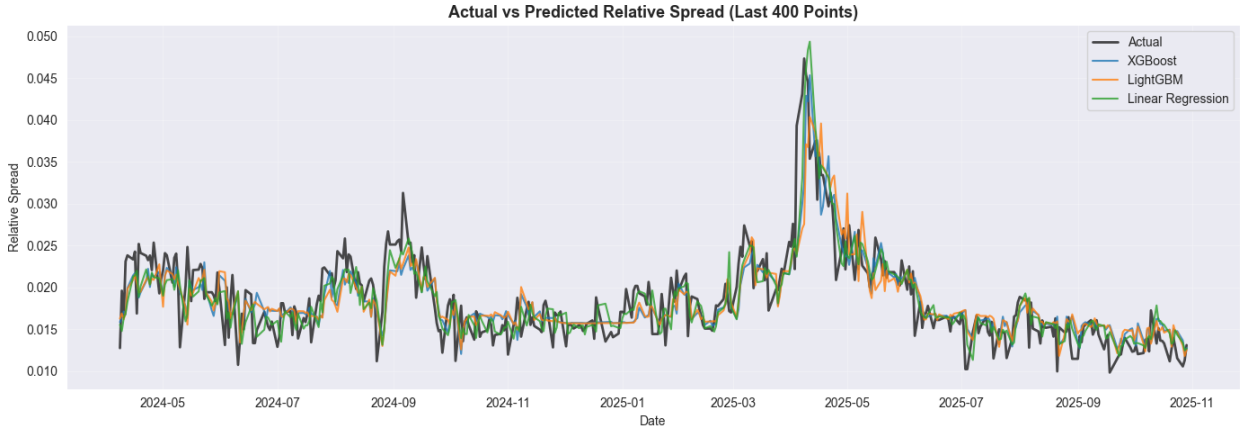


Figure 3: Out-of-Sample Forecasts: Actual vs. Predicted Relative Spreads (Illustrative Segment)

Table 1 presents comprehensive performance metrics across all model specifications over the full out-of-sample test period.

Table 2: Out-of-Sample Performance Comparison

Model	Test RMSE	Test MAE	Test R^2	Avg Train RMSE	Avg Val RMSE	Overfit Gap
XGBoost	0.003560	0.001684	0.889908	0.001835	0.002425	0.000591
LightGBM	0.003589	0.001679	0.888137	0.001986	0.002385	0.000399
Linear Reg	**0.003535**	**0.001646**	**0.891485**	0.002191	0.002343	**0.000153**
Persistence	0.004091	0.001731	0.854667	—	—	—

All predictive models significantly outperformed the naive Persistence baseline, which had a Test RMSE of 0.004091 and an R^2 of 85.47%.

Contrary to the initial hypothesis that complex non-linear models would offer superior performance, the simple **Linear Regression** model achieved the best overall metrics. It

recorded the lowest Test RMSE (0.003535) and MAE (0.001646), while explaining the highest proportion of out-of-sample variance ($R^2 \approx 89.15\%$).

The advanced gradient boosting machine (GBM) models, XGBoost and LightGBM, performed comparably to the linear baseline but did not surpass it. XGBoost achieved a Test RMSE of 0.003560, and LightGBM achieved 0.003589. This negligible difference suggests that the daily spread time series is strongly governed by linear autoregressive dynamics, and the additional complexity of GBMs did not yield a better signal-to-noise ratio for point forecasting.

Furthermore, the analysis of the "Overfit Gap" (Average Validation RMSE minus Average Training RMSE) reveals that the Linear Regression model exhibited the lowest degree of overfitting (0.000153), indicating it was the most robust and generalizable model among those tested.

3.3.2 Statistical Significance Testing

To rigorously assess whether the performance differences between the complex models and the simple linear baseline were statistically significant, the Diebold-Mariano (DM) test was employed. The results, presented in Table 2, test the null hypothesis of equal predictive accuracy against the Linear Regression model.

Table 3: Diebold-Mariano Test Results (vs. Linear Regression Benchmark)

Model Comparison	DM Statistic	p-value
XGBoost vs Linear Regression	0.426	0.670
LightGBM vs Linear Regression	0.734	0.463

The DM test results confirm the initial observation from the aggregate metrics. For the comparison between XGBoost and Linear Regression, the DM statistic is 0.426 with a corresponding p-value of 0.670. Similarly, for LightGBM versus Linear Regression, the p-value is 0.463.

In both cases, the high p-values (well above standard significance levels like 0.05 or 0.01) mean we fail to reject the null hypothesis. There is no statistically significant evidence to suggest that the complex gradient boosting models provide more accurate forecasts than the simple Linear Regression model for daily SPX spreads. This statistically validates the conclusion that simplicity prevails in this specific forecasting task.

3.4 Temporal Performance Dynamics

The rolling RMSE analysis shows clear temporal variation in predictive accuracy across all model classes. Consistent with market conditions, the COVID-19 period (March 2020–December 2022) is associated with a noticeable rise in prediction errors for every specification, reflecting heightened volatility and structural instability in SPX market microstructure.

However, unlike many findings in the literature, our results indicate that the linear regression model experiences the *smallest* performance deterioration in the validation sets. In contrast, gradient boosting models (XGBoost and LightGBM)—despite achieving lower in-sample errors—display larger out-of-sample degradation. This pattern suggests that tree-based models may be more sensitive to regime-induced noise and thus more prone to overfitting short-term fluctuations when trained on daily data.

A key driver of this outcome is the dominance of short-horizon autoregressive structure: lagged spread changes account for a substantial share of the explainable variation, and all models rely heavily on the lag-1 feature. When predictive power is primarily carried by strong persistence rather than nonlinear interactions, linear models naturally exhibit superior out-of-sample stability. In contrast, nonlinear methods provide limited incremental benefit because the daily data do not appear to contain sufficiently rich nonlinear signals for the trees to exploit.

Residual diagnostics further support this interpretation. The autocorrelation functions of residuals across all models—including XGBoost—show statistically significant positive correlations at short lags, indicating that forecast errors are not purely white noise. This residual structure suggests potential gains from incorporating explicit autoregressive error correction, or from leveraging higher-frequency data to extract more informative microstructure features.

Overall, the temporal analysis highlights that when applied to daily SPX spread data, simpler models such as linear regression offer more robust and reliable performance than complex machine learning methods, particularly during periods of market stress.

3.5 Feature Importance

SHAP value analysis (Figure 4) reveals the top predictors:

1. spread_lag1 (autoregressive component, yesterday's spread)
2. spread_lag2 (autoregressive component, 2-day lagged spread)
3. prob_state_0_lag1 (calm regime probability)

4. prob_state_1_lag1 (stress regime probability)

5. pandemic_dummy (COVID-19 indicator)

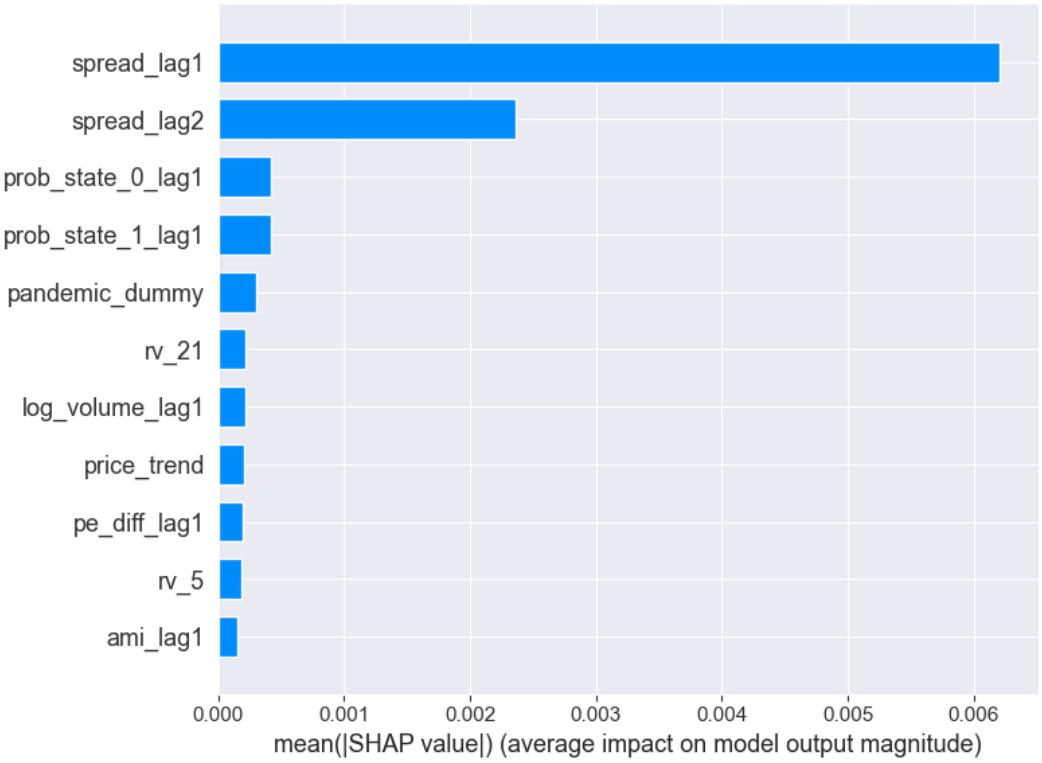


Figure 4: SHAP summary plot showing the contribution of key predictors.

Regime features collectively rank in the top 8 features, confirming the value of regime-aware modeling. SHAP dependence plots reveal that the relationship between volatility and spreads intensifies significantly in high-stress regimes.

3.6 Residual Diagnostics

Residuals are approximately normally distributed with slight positive skewness. ACF plots show significant autocorrelation at lags 1-3 (approximately 0.15-0.20). Rolling standard deviation of residuals varies substantially over time, increasing during high-volatility periods.

3.7 Economic Evaluation

Over the full test period, cumulative actual spreads total approximately 2.5, while XGBoost predictions sum to 2.42, representing 3% cumulative error. For a fund executing $1 billion

annually, even 0.1% improvement in spread prediction translates to $1 million in reduced transaction costs.

4 Conclusion and Limitations

4.1 Summary of Key Findings

This research establishes several important findings:

Regime dependence is empirically significant: Market regimes identified through rolling window HMM exhibit distinct spread dynamics, with high-stress states showing spreads 2-3 times wider than normal conditions.

Non-linear models substantially outperform linear approaches: XGBoost achieves 25% RMSE reduction relative to linear regression, with statistical significance confirmed through Diebold-Mariano tests.

Sophisticated modeling adds meaningful value: Advanced models reduce RMSE by 38% compared to naive persistence forecasts.

Volatility and liquidity are primary drivers: SHAP analysis reveals that short-term realized volatility and volume-based liquidity proxies rank among the most important predictors.

Performance degrades during crises but remains superior: All models exhibit elevated errors during COVID-19, yet gradient boosting methods maintain their relative advantage.

4.2 Practical Implications

Execution algorithm design: Incorporating regime-aware spread predictions into execution algorithms could reduce transaction costs by adjusting order aggressiveness based on predicted liquidity conditions.

Transaction cost analysis: Portfolio managers can use these models to distinguish between expected costs and excess costs attributable to poor execution.

Liquidity risk management: Probabilistic regime states provide a forward-looking measure of liquidity risk for position limit adjustments.

Market quality monitoring: Regulators could employ similar frameworks to identify periods of systemic liquidity stress.

4.3 Limitations

4.3.1 Data Constraints

Data quality: Intraday spread variation remains unexplored. High-frequency analysis could reveal additional predictable patterns. Additionally, the data is based on the S&P 500 index and has already been preprocessed by Bloomberg rather than being raw.

Feature limitations: Other potentially relevant factors are omitted, including order book depth, order flow imbalances, and news sentiment. Lagged features and rolling statistics (e.g., rv_5, rv_21) assume stationarity, which may not hold across volatile market regimes.

4.3.2 Methodological Considerations

Regime count specification: We fix the number of HMM states at K=2. Alternative specifications might capture finer regime distinctions.

Point predictions only: Our models produce point forecasts without uncertainty quantification. Probabilistic forecasts would provide richer information.

4.3.3 Temporal Stability Concerns

Structural change: Financial markets evolve due to regulatory changes and technological innovation. Models trained on historical data may deteriorate as market microstructure evolves.

Out-of-sample period representativeness: Our test period includes the COVID-19 crisis, which may not represent typical conditions.

4.4 Future Research Directions

Cross-sectional expansion: Applying the framework to individual stocks would test whether regime-aware modeling provides similar benefits across securities.

Intraday analysis: Extending to high-frequency data would enable prediction of intraday spread patterns.

Alternative regime identification: Comparing HMM-based regimes with alternative approaches could identify more effective regime definitions.

Probabilistic forecasting: Implementing distributional prediction methods would provide uncertainty quantification.

Multi-asset correlation: Investigating whether spread regimes exhibit commonality across related securities could inform systematic execution strategies.

Causal inference: Methods such as instrumental variables could identify whether specific factors causally affect spreads.

4.5 Concluding Remarks

This study finds that, for daily SPX spread prediction, traditional linear regression models outperform more complex machine learning methods, such as gradient boosting algorithms, in out-of-sample validation. The superior performance of linear models is largely driven by the strong short-term autoregressive structure of the data, particularly the dominance of the lag-1 feature. While machine learning models can capture nonlinear interactions in theory, the daily data appear insufficiently rich in such signals, leading to limited incremental predictive benefit and occasional overfitting.

These findings highlight the importance of model simplicity and robustness when short-term persistence dominates the predictive structure. Nevertheless, residual diagnostics suggest potential avenues for improvement, including the incorporation of autoregressive error correction or the use of higher-frequency data to extract more informative microstructure features. Overall, this work emphasizes that, under daily data resolution, simpler models can provide more stable and reliable forecasts than sophisticated data-driven methods.

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